

DATA ANALYTICS PROJECT

STUDENTS ACADEMIC PERFORMANCE

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We use tech to connect human potential and
opportunity with dignity & humility

Project description

Students Academic Performance Dataset

- Uses of the data set(educational analytics(predicting the final grades and the exam outcome),Early Intervention(academic support from parents),Research(exploring learning behaviour)
- Project goals(predict students final grades,analyze factors of the predictive success,determine how various features affect performance,analyze trends in different performance groups)
- Solutions(data processing and cleaning, data analysis)
- Why(educational impact,data science learning,research)



Data cleaning



- Data Importing(`pandas.read_csv()` in Python, importing the CSV dataset)
- Understanding the dataset(Filling missing values (mean, median, mode),Dropping rows or columns with too many missing values,**Purpose:** Ensure no empty or incomplete data affects analysis)
- Removing Duplicates(using `df.drop_duplicates()`),**Purpose:** Ensure no empty or incomplete data affects analysis)
- Correcting data types(Converting text numbers to integers,Converting dates to proper date formats,Changing categorical variables to category type, Converting text numbers to integers,Converting dates to proper date formats, Changing categorical variables to category types)
- Saving cleaned dataset(`df.to_csv("cleaned_student_data.csv", index=False)`)

Data Analysing

- Exploratory Data Analysis (EDA)(Plot grade distributions,Identify factors that affect academic performance,Compare performance among different groups of students)
- Feature Selection(Which features strongly influence grades,Which ones are weak or irrelevant)
- Graphs and charts
- Python



Thank you!

